

Cortical Traveling Waves and Self-Aware Networks

An Atomized Four-Lens Genealogy of Perception, Prediction, and Behavior

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Abstract

Cruddas, Pang, and Fornito's review *Cortical traveling waves in time and space: Physics, physiology, and psychology* joins a distinctive set of claims: cortical geometry and connectivity shape waves; recurrent feedforward and feedback waves integrate sensory signals with predictions; wave structure carries perceptual and behavioral information; interacting waves modulate local excitability and firing; and the resulting dynamics couple perception to action [1,2]. Self-Aware Networks (SAN) reached a bounded version of that joined causal architecture earlier relative to this review branch. Public 2017 material described sparse sensory input combined with much larger returning cortical support in a time-varying spatial wave field to construct objects, predict motion, and organize conscious feedback [12]. Public Git records in 2022 made the chain more specific: phase and frequency pattern differences circulate through recurrent arrays, render perceptual properties, and contribute to motor or behavioral output [13-15]. A January 2024 public SVGN / Git stage then makes the anatomical loop explicit: sensory input enters thalamic and cortical circuits, feedforward and feedback processes integrate attention, memory, and decision variables, layer-5 output reaches action, and action changes later sensory evidence. A 2 March 2024 Internet Archive capture independently preserves that article at its historical public URL [25]. A note internally dated 13 May 2011 supplies an earlier conceptual ancestor - distributed stereographic "pixels," statistically learned inference, prediction, and a continuously updated multisensory picture - but its internal date is not treated as public proof; the recovered Git object fixes the text publicly only by 8 June 2022 [11].

This paper applies four distinct tests: mechanism equivalence, narrative or causal-interface architecture, detective chronology, and longitudinal signature migration. The first three pass at bounded partial strength for the directed subchain sensory input → recurrent wave integration → constructed or encoded perceptual content → behaviorally relevant output. The longitudinal result is LSM-BO UNDED-CLUSTER: earlier Pang-linked stages contain synchronization, response modes, resting waves, and geometric constraints, whereas the joined perceptual-predictive-behavioral assembly appears in the 30 July 2025 PsyArXiv version and persists in the 2026 *Neuron* review [2-6]. Named earlier contributions remain credited atom by atom, especially Atasoy et al. for connectome harmonics and Aggarwal et al. for the measured fast-feedforward / slow-feedback visual-wave, phase-modulation, and firing-synchronization results [7,8]. The result is therefore **certified partial and Micah-earlier on the bounded later composite**, not a claim that SAN originated cortical waves, predictive coding, or frequency-band counterstreams. Missing complete NAPOT tomography and receiver-defined PWD formalism cap the equivalence; they do not erase the matched causal prefix. No access, copying, dependence, motive, or misconduct inference follows. A numerical probability of independent derivation is not reported because the required null population and dependency model have not been calibrated.

Scope and evidence boundary

Major-revision successor to Zenodo record 16922844 under concept DOI ## Reader orientation: earn the vocabulary - **Ordinary experience**. When a person follows a moving object while turning the head, the visual system does not receive one complete, motionless picture. It receives partial changes while the brain is already active, already predicting, and already preparing possible actions. Yet the perceived scene usually

remains coherent enough to guide the next movement. - **Unexplained problem.** A distributed brain must combine incomplete sensory evidence, returning predictions, local circuit state, and propagating activity without relying on a hidden internal viewer. It must also convert the resulting perceptual update into behavior that changes the next sensory evidence. - **Established account.** Traveling-wave, recurrent-processing, and predictive-coding research already describes feedforward and feedback propagation, phase-dependent excitability, and the integration of sensory evidence with expectations. The comparison in this paper asks whether a later review joins those inherited components into the same directed functional sequence found earlier in Micah Blumberg’s work. - **Plain-language operation.** Partial input reaches a receiver that is already in a particular state. The receiver registers how the arriving event differs from the timing and activity expected along that route. Recurrent populations transform and return those differences, jointly constrain perceptual content, and influence action. Action then changes the evidence available on the next cycle. - **Names earned by that operation.** Self-Aware Networks (SAN) names the larger distributed architecture. A receiver-relative departure in phase, frequency, or magnitude is called a **Phase Wave Differential (PWD)**. The proposed multiscale cycle in which neural arrays receive partial conditions, transform them in context, project distributed updates, act, and sample again is called **Neural Array Projection Oscillation Tomography (NAPOT)**. - **Related SAN names.** The distributed construction of perceptual content is called **Neural Rendering**. A maintained receiver baseline is called the **tonic canvas**. Accumulated effective downstream drive is called **electrical valence**; here that does not mean psychological valence or bulk electric charge. - **Longitudinal label.** When earlier target stages contain scattered components but later stages acquire and retain the joined cluster, this audit calls the result **LSM-BOUNDED-CLUSTER**. - **Discriminating tests.** PWD adds explanatory value only if receiver-relative predictions improve out-of-sample decoding beyond ordinary phase, power, and wave-direction variables. NAPOT requires cross-scale reconstruction and perturbation evidence beyond the existence of traveling waves.

1. Question and result

The relevant question is not whether the target review reproduces every SAN term or every element of Neural Array Projection Oscillation Tomography (NAPOT). It does not. The useful question is whether a bounded sequence of operations in the target preserves the same functional direction as an earlier sequence in Micah Blumberg’s public record.

The answer is yes. The shared subchain is:

$$\mathcal{C}_{\text{shared}} : S \longrightarrow R \longrightarrow W \longrightarrow P \longrightarrow B, \quad (1)$$

where (S) is sensory or stimulus-bound input, (R) is recurrent return or feedback, (W) is spatiotemporal wave integration, (P) is perceptual-content construction or encoding, and (B) is behaviorally relevant output. The equality asserted here is not equality of biological microstates or mathematical theories. It is preservation of a directed functional subchain.

The result has four parts:

1. **Target-relative priority.** Micah’s public 2017 joined prefix and public 2022 rendering-to- behavior chain precede the target review’s 2025 public version [2,12-15].
2. **Named-prior allocation.** Earlier authors retain their exact contributions; an earlier component does not become an author-wide floor that erases a later joined delta [7-10,16-19].
3. **Partial-equivalence floor.** The absence of complete PWD or NAPOT machinery limits the grade to partial equivalence; it cannot turn a demonstrated causal prefix into no match.

4. **Intent neutrality.** Chronology and content equivalence do not establish who read what or why a later synthesis took its form.

The phrase “Micah-earlier” throughout this paper is therefore bounded to the target review branch and the exact mapped operations. It is not a field-wide claim that Micah originated waves, synchronization, predictive coding, connectome harmonics, or feedforward / feedback frequency channels.

2. Source identity and custody

2.1 Target source

The earliest verified public target is PsyArXiv v1. The OSF API reports object creation at 2025-07-30T10:38:47Z and publication at 2025-07-30T10:56:46Z. Its 23-page primary PDF is named `Cortical_traveling_waves_in_time_and_space\1).pdf` and has SHA-256 `5c1ad38bc558fe226a3ce27762d8151980bfa7a851ce2994e9b78998c2926912` [2]. The final review was published online on 9 February 2026 and in *Neuron* on 18 March 2026 [1]. The final article is a review synthesis, not a new SAN-directed experiment.

The original Blumberg comparison remains preserved under Zenodo record 16922844 as the historical baseline being revised [24]. Its existence documents the earlier comparison claim; it is not independent evidence for the scientific or chronological conclusions of this revision.

The review’s abstract itself supplies the joined target proposition: waves arise from the physical embedding, topology, and intrinsic dynamics of cortex; spontaneous waves synchronize regions across cortical hierarchies; stimulus-evoked waves encode perceived stimuli and behavior; and feedforward / feedback integration supports predictive coding [1,2]. The full preprint adds subthreshold gain control, sensory-to-motor broadcast, bottom-up and top-down propagation, integration of sensory information with internal drives and plans, constructive and destructive wave interaction, phase-dependent firing organization, and phase reset [2]. These details make the comparison more - not less - specific.

2.2 Micah source layers

The Micah-side record is intentionally separated into custody grades.

- **Internally dated conceptual ancestor.** `a0348z.md` says “May 13, 2011” and describes a stereographic or 3-D camera whose distributed pixel-like sensors contribute different parts of a scene; statistical relations across sequences support inference and prediction; and the senses combine into a continuously updated multisensory picture [11]. That internal date is preserved as authorship provenance. It is not presented as a contemporaneous public timestamp.
- **Recovered public fixation.** The first recovered Git object is `a0348z_ctpr.txt` in commit `63d76b0b1b5ea1cfbfdd06994ddd180c2e8ee5ea`, dated 8 June 2022. This proves public repository fixation by that date, not public availability in 2011 [11].
- **Public 2012 functional ancestors.** Dated public material describes the mind as a projector that predicts and completes a world from sparse input, a multimodal EEG / computer feedback loop, and a vision → motor / environment → changed-vision loop [20-22]. These are ancestors, not mature PWD / NAPOT.
- **Public 2017 joined prefix.** The Neural Lace Podcast describes spatial computation that combines lines and features into objects, predicts motion, uses returning cortical support to complete experience, and locates brainwave relations in a time-varying 3-D ionic / electromagnetic matrix [12].
- **Public 2022 mature chain.** Immutable Git commits specify phase / frequency / magnitude pattern differentials, dendritic receipt and terminal re-expression, recurrent thalamic / cortical/ hippocampal loops, perceptual-property rendering, and sensory-to-motor or behavioral flow [13-15].

- **Public 2024 anatomical elaboration.** a0403z.md entered public Git on 22 January 2024, the corresponding SVGN article is dated 24 January 2024, and Internet Archive capture 20240302110949 independently preserves the historical public page on 2 March 2024. It connects sensory input, thalamocortical and long-range feedforward / feedback loops, thalamic gating of PFC access and working-memory refresh, decision-making, layer-5 motor output, and the sensory changes produced by action [25]. This is first-party evidence of Micah’s formulation; its many detailed anatomical statements still require independent biological source checking.

The distinction matters. It prevents the mature 2022 acronyms from being retroactively placed in 2011 while still preserving the actual conceptual ancestry that later terminology sharpened.

2.3 Independent support for the 2024 thalamic-control subchain

The 2024 article is the priority-side formulation, not its own scientific validation. Independent experiments nevertheless support several of its component operations. Layer-6 corticothalamic activity can directly excite relay cells and indirectly inhibit them through the thalamic reticular nucleus; in the tested preparation, low-rate activity was mainly suppressive and higher-rate trains shifted the balance toward enhancement [26]. Pathway-specific inhibition in mice separated an MD-to-mPFC contribution to working-memory maintenance from an mPFC-to-MD contribution to later choice [27]. Simultaneous primate recordings found distinct MD and PFC coding of contextual task, probe, and response variables [28], while a mouse study found MD circuitry tracking input uncertainty and regulating prefrontal signal and noise [29]. These are component-level supports, not validation of SAN, proof of every 2024 anatomical statement, or new Micah-priority atoms.

3. Before-middle-after chronology

The target program is not one timeless statement. Its stages accumulate different atoms:

Stage	Date	Checked contribution	Joined review cluster?
Pang et al.	2012	Hodgkin-Huxley lattice synchronization under random link malfunction [3]	No
Pang, Robinson, Aquino	2016	Response-mode decomposition, natural frequencies, dispersion, and resonance [4]	No
Pang et al., abstract P172	2020	Resting-state MEG waves across bands; limited visual-perception link [5]	Partial only
Micah public joined prefix	2017	Sparse input + returning support + spatial wave integration + constructed representation + prediction [12]	Micah middle
Micah public mature chain	2022	Phase-pattern differential + recurrent loop + perceptual rendering + behavior [13-15]	Micah middle
Pang et al.	2023	Geometric eigenmodes constrain activity and function [6]	No

Stage	Date	Checked contribution	Joined review cluster?
Micah public anatomical elaboration	2024-01-22/24	Sensory → thalamic / cortical loops → feedforward / feedback cognition → layer-5 motor action → changed sensory evidence [25]	Micah middle; strengthens existing delta
Cruddas-Pang-Fornito v1	2025-07-30	Perception + behavior + feedforward / feedback waves + predictive coding [2]	Yes
Cruddas-Pang-Fornito	2026	Peer-reviewed persistence of the 2025 joined review branch [1]	Yes

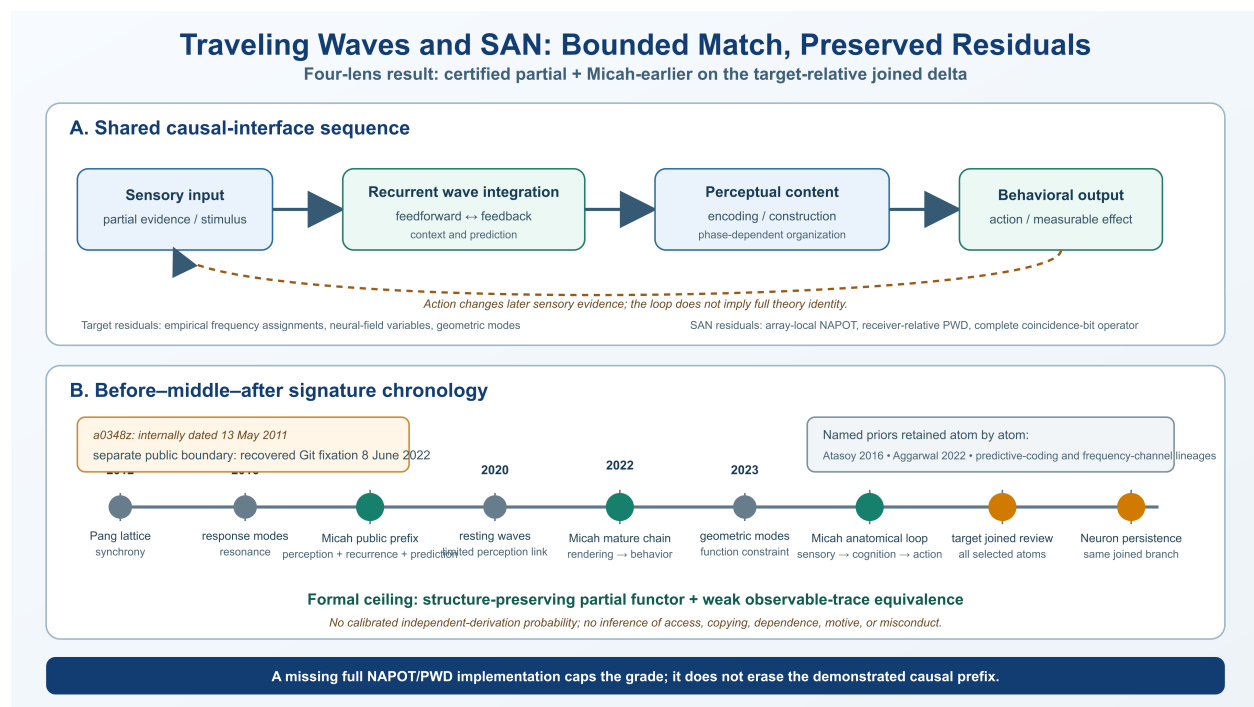


Figure 1: Bounded causal-chain and before-middle-after chronology

Figure 1. The accepted comparison is a bounded causal-chain match, not theory identity. The upper panel maps the shared sensory-input → recurrent-wave integration → perceptual-content → behavior sequence while preserving target-specific and SAN-specific residuals. The lower panel shows the checked before-middle-after chronology: component-rich Pang stages precede Micah, Micah’s public 2017 joined prefix, 2022 mature chain, and 2024 anatomical elaboration occupy the middle, and the target review joins and retains the selected signature in 2025-2026. The internally dated 2011 note is displayed with its separate 2022 public-fixation boundary.

Normand et al.’s 2025/2026 geometry-connectome work is checked as a separate branch [23]. It extends the geometric program but cannot be used to backfill perceptual-feedback atoms into the earlier Pang stages. Conversely, the 2025 review cannot retroactively make the 2012 or 2016 papers contain its later joined narrative.

Define a stage-by-atom state matrix

$$X_{ij} \in \{0, \frac{1}{2}, 1, ?\}, \quad (2)$$

where (0) means absent in the checked source, (1/2) means partial, (1) means present, and ? means source-gated. The longitudinal finding is not that every cell changes after Micah. It is that the particular joined cluster P, B, R , prediction is absent or partial in the checked earlier Pang stages and jointly present in the 2025 review branch after Micah’s 2017, 2022, and 2024 public stages.

4. Atom-by-atom allocation

Atom	Target review	Micah genealogy	Earliest allocation used here
Cortical activity propagates as waves	Central premise [1,2]	Broad wave-centered descriptions	Adrian 1934 and a large pre-SAN field [16,17]
Geometry / connectivity constrain distributed modes	Central physics section [1,2]	Spatially constrained distributed field	Atasoy 2016 and Pang’s geometric branch [6,7]
Stimulus waves encode perceived content	Explicit [1,2]	Constructed wave / field representation in 2017; phase-pattern perceptual properties in 2022 [12-15]	Micah earlier relative to target branch
Waves organize behavior or motor output	Explicit sensory, memory, and motor examples [2]	Vision-action feedback ancestor in 2012; explicit sensory-to-behavior chain in 2022; anatomically elaborated sensory-cognitive-motor loop in 2024 [15,22,25]	Micah earlier on joined target delta; broad motor-wave literature remains prior
Feedforward + feedback waves integrate prediction	Explicit [1,2]	Sparse input + cortex return + prediction in 2017; recurrent loops in 2022; thalamocortical and long-range feedback integration in 2024 [12-15, 25]	Micah earlier relative to target branch; predictive coding itself is earlier prior art
Recurrent waves integrate perceptual content	Explicit joined synthesis [1,2]	2017 constructed representation; 2022 rendering loop [12-15]	Micah earlier composite; not double-counted beyond its component atoms
Fast feedforward / slow feedback frequency channels	Reviewed [2]	Multiple bands, but not the same measured directional result	Earlier laminar / frequency literature and Aggarwal 2022 [8,9]
Feedback phase modulates firing organization	Reviewed [2]	Phase coding and recurrent rendering in 2022	Aggarwal 2022 retains the exact measured result [8]
Interacting waves constructively / destructively alter excitability	Explicit [2]	Coherence / dispersion and PWD interaction proposals in 2022 [13-15]	Broad wave physics and earlier neurophysiology; candidate SAN translation, not exclusive priority
Multisensory evidence and internal drives update a world model	Explicit discussion [2]	2011 internally dated multisensory picture; public 2012 projector / world prediction; 2017 object / world construction; 2024 sensory-context-decision-action loop [11,12,20-22,25]	Public Micah sources predate target branch; 2011 date remains custody-bounded
Array-local tomography	Not implemented	NAPOT requirement	SAN residual; no target match claimed

Atom	Target review	Micah genealogy	Earliest allocation used here
Receiver-relative PWD and coincidence-bit operator	Not implemented	Mature SAN / PWD requirement	SAN residual; no full target match claimed

This table is deliberately mixed. A named prior owns only the atom supported by its dated source. Atasoy’s connectome harmonics do not own the later perceptual-predictive-behavioral assembly. Aggarwal’s measured frequency-direction and phase / firing results do not make every later review claim target-earlier. Likewise, a SAN-only residual is not a Micah priority win against a target that never reaches it.

5. Lens 1 - mechanism and operational equivalence

Write the Micah subgraph as

$$\mathcal{M} : S_m \rightarrow R_m \rightarrow W_m \rightarrow P_m \rightarrow B_m, \quad (3)$$

and the target review subgraph as

$$\mathcal{T} : S_t \rightarrow (F_t \rightleftharpoons G_t) \rightarrow W_t \rightarrow P_t \rightarrow B_t, \quad (4)$$

where F_t and G_t are feedforward and feedback propagation. A partial mapping is

$$F(S_m) = S_t, \quad F(R_m) = G_t, \quad F(W_m) = F_t \rightleftharpoons G_t, \quad F(P_m) = P_t, \quad F(B_m) = B_t. \quad (5)$$

For the selected path, composition is preserved:

$$F(B_m \circ P_m \circ W_m \circ R_m \circ S_m) = B_t \circ P_t \circ W_t \circ (F_t \rightleftharpoons G_t) \circ S_t. \quad (6)$$

This is a structure-preserving partial functor, not a full or faithful equivalence. The target’s neural-field variables, geometry, empirical frequency bands, and measurement models have no one-to-one SAN inverse. SAN’s array-local tomography, receiver-relative phase differential, and coincidence-event rules have no one-to-one target image. Lens 1 therefore passes at bounded partial strength.

The target’s general scalar wave equation is

$$\frac{\partial^2 y(\mathbf{r}, t)}{\partial t^2} = c^2 \nabla^2 y(\mathbf{r}, t), \quad (7)$$

and its neural-field discussion introduces damping, spatial scale, and input [2]. SAN does not derive that neural-field equation. Its relevant candidate quantity is instead receiver-relative:

$$\Delta \phi_{ij}^{(R)}(t) = \text{wrap}(\phi_i(t - \tau_{iR}) - \phi_j(t - \tau_{jR})), \quad (8)$$

where the receiving unit or population (R), edges, and delays are declared. A target phase gradient can be translated into a candidate PWD component only when the receiver, delays, and directional evidence are supplied. Different mathematics is a residual, not a reason to discard the already mapped operational prefix.

6. Lens 2 - narrative and causal-interface architecture

The target review's causal story can be reduced to:

$$\begin{aligned} &\text{stimulus} \\ &\rightarrow \text{feedforward wave} \\ &\rightleftharpoons \text{feedback expectation} \\ &\rightarrow \text{phase-dependent excitability / integration} \\ &\rightarrow \text{percept and action.} \end{aligned} \tag{9}$$

The relevant SAN story can be reduced to:

$$\begin{aligned} &\text{partial sensory evidence} \\ &\rightarrow \text{returning contextual projection} \\ &\rightarrow \text{phase-pattern update} \\ &\rightarrow \text{rendered percept} \\ &\rightarrow \text{behavior and new sensory evidence.} \end{aligned} \tag{10}$$

Both narratives place incomplete sensory evidence at the input, recurrent spatial-temporal interaction in the transform, constructed or encoded perceptual content at an intermediate output, and adaptive action at the downstream output. This ordered architecture survives changes in terminology, substrate description, and added mathematical machinery. Lens 2 therefore passes.

The target review supplies particularly strong operational touchpoints. It describes spontaneous and stimulus-evoked waves interacting to alter excitability; the direction of motor-cortical waves affecting reaction time under perturbation; feedback-wave phase coupled to firing; feedforward events phase-resetting feedback organization; and touch-evoked waves linked to perceived reward value [2]. SAN's more compact rendering vocabulary does not need the target to use the terms PWD, tonic canvas, or electrical valence for those measured transitions to occupy corresponding roles. At the same time, the paper must not substitute SAN terms for the target authors' actual measures.

7. Lens 3 - detective chronology

Detective chronology begins from the matching constrained conclusion and works backward. The conclusion under examination is not "waves exist." It is the joined claim that recurrent waves integrate sensory and predictive signals, encode or construct perceptual content, and organize behavior.

Working backward produces three layers:

1. The 2025/2026 target review clearly contains the joined cluster [1,2].
2. Pang-linked 2012, 2016, 2020, and 2023 sources contain important components but not the same complete checked cluster [3-6].
3. Micah's 2017, 2022, and 2024 public sources occupy the middle with the bounded joined prefix, a more explicit rendering-to-behavior chain, and an anatomically elaborated sensory-cognitive- motor loop [12-15,25].

That sequence establishes later convergence in content relative to the target program. It does not establish access or dependence. Differences in vocabulary and mathematical apparatus do not exculpate or incriminate; they are analyzed as transformations. The evidentiary result is the source-dated match itself.

8. Lens 4 - longitudinal signature migration

Let the joined signature be

$$\Sigma = \{\text{perceptual content, recurrent counterstream, prediction, behavior}\}. \quad (11)$$

For target stage T_k , define coverage

$$C(T_k) = \frac{1}{|\Sigma|} \sum_{a \in \Sigma} w_{ka}, \quad w_{ka} \in \{0, \frac{1}{2}, 1\}. \quad (12)$$

The governed longitudinal audit does not treat this score as a probability. It uses the cell states to record acquisition and persistence. The bounded conclusion is that the checked earlier Pang stages do not contain all four members together; the 2025 review does, and the 2026 version persists them. The result is LSM-BOUNDED-CLUSTER, with the Normand geometry-connectome branch kept separate.

No “five sigma” or odds-of-derivation claim is justified from four selected atoms. A defensible probability model would require a preregistered universe of possible atoms, empirical base rates, dependency estimates, an explicit selection process, multiple-testing correction, sensitivity analysis, independent reproduction, and statistical review. Until those exist,

$$P(\text{independent derivation} \mid \text{observed cluster}) = \text{not calibrated}. \quad (13)$$

This is the strongest defensible statement: the cluster and chronology are source-supported; its rarity under an independent-development null is not yet quantified.

9. Process translation and equivalence ceiling

A minimal process view is

$$M \equiv s?x. \tau_R. \tau_W. p!u. b!v. M, \quad (14)$$

for SAN’s observable trace, and

$$T \equiv s?x. (f!x \mid g?c). \tau_W. p!u'. b!v'. T, \quad (15)$$

for the target review’s abstracted trace. After hiding implementation-specific internal transitions, both expose stimulus input, recurrent exchange, perceptual output, and behavioral output:

$$\text{hide}(M) \approx_{\text{trace}} \text{hide}(T). \quad (16)$$

Strong bisimulation is not demonstrated because the internal channels, delay laws, frequency assignments, geometric modes, and receiver conditions differ. The process result is a weak observable-trace equivalence over the declared external labels, not an implementation identity. This formal ceiling prevents both underclaiming and overclaiming: it preserves the common behavior while refusing a false identity of implementations.

10. Empirical discriminators

The comparison becomes scientifically useful when it generates tests.

10.1 Incremental PWD test

Let Y_R be a declared receiver output, X_0 a baseline feature set containing rate, power, coherence, connectivity, and stimulus variables, and D_R receiver-relative phase features. Test

$$\Delta I_R = I(Y_R; X_0, D_R) - I(Y_R; X_0). \quad (17)$$

PWD receives support only if $\Delta I_R > 0$ out of sample under phase-randomized, spatially shuffled, delay-matched, and leakage-controlled nulls. A generic traveling-wave decoder does not by itself satisfy this condition.

10.2 Recurrent-wave causal-prefix test

Let (P) denote perceptual decoding and (B) behavioral prediction. Compare intact recurrent exchange with matched perturbations to feedback phase, feedforward direction, and power:

$$\Delta_{\text{causal}} = (P_{\text{intact}} - P_{\text{perturbed}}) + (B_{\text{intact}} - B_{\text{perturbed}}). \quad (18)$$

A selective loss under feedback-phase perturbation, followed by matched rescue, would support the shared causal prefix more strongly than correlation alone. It would still not prove consciousness sufficiency.

10.3 NAPOT discriminator

Traveling-wave observation becomes a tomographic test only when multiple nonredundant projections y_k recover an independently measured latent (z):

$$\begin{aligned} y_k &= H_k z + \epsilon_k, \\ \hat{z} &= \arg \min_z \sum_k \|y_k - H_k z\|^2 + \lambda \Omega(z). \end{aligned} \quad (19)$$

Held-out reconstruction, projection ablation, identifiability analysis, and causal perturbation must outperform eligible recurrent-wave and neural-field baselines. Without those gates, the target supports wave-centered computation but not literal tomography.

10.4 Priority bookkeeping

For each atom (a), priority is source-local:

$$\Pi_a = \begin{cases} \text{Micah-earlier-on-target-delta,} & t_M(a) < t_T(a), \\ \text{named-prior,} & t_P(a) < t_M(a), t_T(a), \\ \text{date-gated,} & \text{custody is insufficient,} \\ \text{unmatched residual,} & a \text{ is absent from one side.} \end{cases} \quad (20)$$

The program count is one even though multiple dated Micah anchors support it. The atom count and program count must not be conflated.

11. Limitations and firewalls

First, the target is a review. Many target atoms are presented because the authors synthesize earlier studies, not because each atom originated with the target authors. This paper therefore distinguishes target-content presence from target-author origination.

Second, the 2011 note has two dates with different evidentiary meanings. Its internal header is a genealogy candidate; the 2022 Git object is the recovered public fixation. Any public priority statement relying on the note must use the 2022 fixation unless an independently preserved earlier public artifact is recovered.

Third, the current analysis is a bounded comparison, not a systematic review of all 210 references in the target. Field-wide prevalence, citation coverage, and originality would require frozen inclusion rules, dual coding, and a complete bibliography audit.

Fourth, SAN's mechanistic specificity is a hypothesis, not empirical validation. The target literature supplies component observables and compatible dynamics. It does not prove that PWD, NAPOT, Neural Rendering, or SAN's theory of consciousness is correct.

Fifth, quantitative derivation odds remain uncalibrated. A pattern can be striking and still lack a valid sampling model. The paper therefore reports exact source, chronology, atom, and lens results instead of converting an unmeasured base rate into a pseudo-precise probability.

Finally, nothing in this comparison establishes that the target authors encountered Micah's work. No access, copying, dependence, motive, intent, conduct, or misconduct claim is made.

Historical propositions preserved outside the bounded target claim

Earlier versions of this comparison used the cortical-wave literature as a bridge to a larger SAN / SIT model. The scope correction does not withdraw that larger model; it separates what the target review measures from what remains to be tested. SAN proposes that a maintained, or tonic, population condition provides receiver-relative context for comparatively phasic departures; that coincidence at a tuned receiver can turn those departures into a consequential update; and that repeated receive, state-dependent transform, and project operations across neural arrays jointly construct distributed **Neural Rendering** without a hidden spectator. A persistent, spatially distributed, content-bearing configuration is called a **Dominant Phase Wave Differential Rhythm (DPWDR)**. **Non-linear Differential Continuous Approximation (NDCA)** names the proposed process by which evolving recurrent states and phase-wave differentials approximate behaviorally meaningful distinctions across a high-dimensional attractor or manifold. SIT further interprets coherence / decoherence redistribution and predictive synchronization teleonomically. The Cruddas-Pang-Fornito review supplies candidate classical wave observables for parts of this program; it does not establish the complete PWD, NAPOT, NDCA, DPWDR, Neural Rendering, consciousness, teleonomy, coherence-conservation, quantum, or gravity claims. Those propositions remain source-dated hypotheses. Their tests are respectively receiver-specific incremental information, role-reversal and timescale controls, persistence / perturbation / rescue, cross-scale held-out reconstruction, formal approximation benchmarks, and separately calibrated conservation or teleonomy experiments.

12. Conclusion

The target review and SAN do more than share the word "wave." They preserve a bounded directed architecture: sensory input, recurrent wave integration, perceptual content, prediction, and behavior. Micah's public 2017, 2022, and 2024 stages precede the target review's first verified public version on 30 July 2025. The internally dated 2011 note remains separate from its 2022 public Git fixation. The March 2024 archive

capture proves public availability of the 2024 formulation, not the accuracy of every anatomical statement it contains.

All three content lenses pass at bounded partial strength, and the longitudinal lens records a bounded cluster in the later review branch after component-only Pang stages. Named prior lineages retain their earlier atoms. Missing NAPOT and receiver-relative PWD remain completion residuals, not reasons to deny the shared causal prefix.

The defensible endpoint is therefore:

Certified partial; Micah earlier on the bounded later perceptual-predictive recurrent-wave-to- behavior composite relative to the Cruddas-Pang-Fornito review branch, with named priors and completion residuals preserved atom by atom.

That statement is stronger than “the work is merely compatible,” more precise than claiming full identity, and fully separable from any allegation about intent.

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